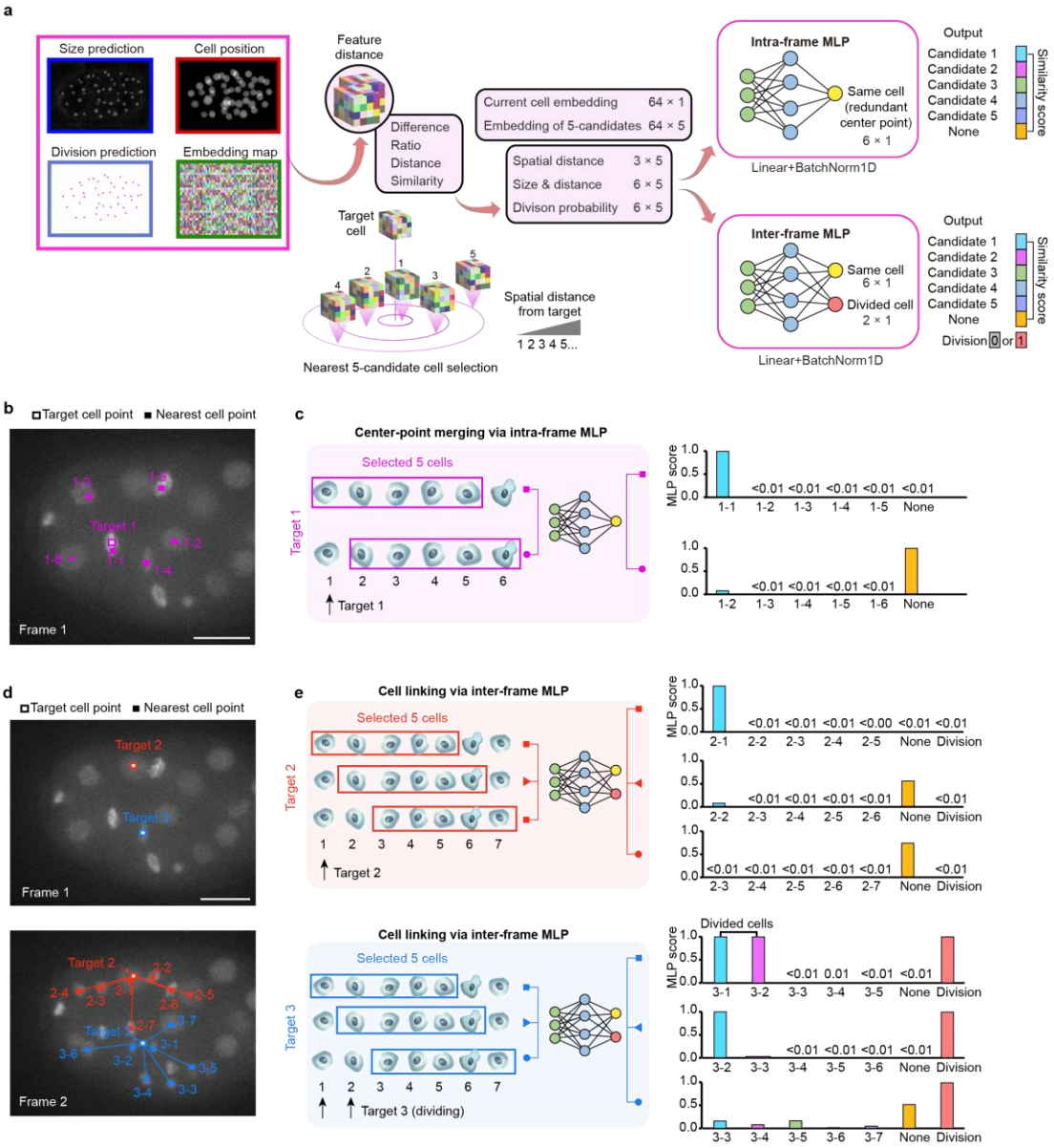


Supplementary Figure 5 | Illustrations of the intra-frame MLP and inter-frame MLP. a, Illustration of the workflow of 2 MLPs. The 5 nearest candidate cells to the target cell are selected based on the spatial distances between cell centers. For intra-frame MLP, we selected the 5 candidate cells within the same time frame. For inter-frame MLP, we selected the 5 candidate cells in the next frame closest to the target cell in the current frame. For each candidate, four feature distances relative to the target cell are computed: (1) size and division probability difference, (2) size and division probability ratio, (3) Euclidean distance, and (4) feature similarity. These metrics serve as inputs to the MLPs. The intra-frame MLP identifies redundant cell center point detections of the same cell, while the inter-frame MLP determines whether the target cell matches a candidate in the next frame or undergoes division. Each MLP outputs 6 similarity scores including 5 candidates and one 'None' class indicating that the target cell is not matched with any of the 5 candidates. The inter-frame MLP outputs 6 similarity scores and also provides a division prediction. **b**, Representative example of intra-frame MLP. For a given target cell (target 1), its six nearest cell center points (labeled 1-1 to 1-6) are identified within the same frame. We note that point 1-1 is a redundant detection of the same cell as the target. **c**, Intra-frame MLP outputs for different combinations of 5